📘 Class 12 IP | Computer Networks

Part 1: Introduction to Computer Networks

1. Introduction to Networks

> **Definition:** A **computer network** is a collection of interconnected computers and other devices (like printers, scanners) that can communicate with each other and share resources (like files, printers, and internet connection).

Key Advantages of a Network:

- **Resource Sharing:** Share hardware (printers, scanners), software, and data.
- **Communication:** Email, instant messaging, video conferencing.
- **Data Centralization:** Store data on a central server for easy access and backup.
- **Cost Efficiency:** Reduce costs by sharing expensive resources.
- **Reliability:** If one computer fails, others can take over its work.

2. Types of Network: PAN, LAN, MAN, WAN

Networks are classified based on their geographical area (range):

- **PAN (Personal Area Network):** A network organized around an individual person. Range is typically up to 10 meters. (e.g., Bluetooth connection between a phone and a headset).
- **LAN (Local Area Network):** A network that connects devices within a small, limited area like a home, office, or school. (e.g., your home Wi-Fi network).
- **MAN (Metropolitan Area Network):** A network that spans a city or a large campus. It is larger than a LAN but smaller than a WAN. (e.g., a cable TV network in a city).
- **WAN (Wide Area Network):** A network that spans a large geographical area, such as a country, continent, or the entire world. The Internet is the largest WAN. (e.g., a bank's network connecting its branches across the country).

Range, Examples, Use Cases

Network Type	Full Form	Geographical Area Covered	Examples	Use Case
PAN	Personal Area Network	Very small area around a person (Up to 10 m)	Bluetooth headset, smartwatch, wireless keyboard, NFC payment	Connecting personal devices around one person
LAN	Local Area Network	Typically a building or a campus (Up to 1 km)	Home Wi-Fi, school computer lab, office network, college campus	Sharing files, printers, and internet within a small area
MAN	Metropolitan Area Network	A city or a metropolitan area or a large campus (10 km to 100 km)	Cable TV network, city-wide Wi-Fi, bank branches in a city, metro rail network	Connecting multiple LANs across a city
WAN	Wide Area Network	Large geographical area — countries or worldwide (more than 100 kms)	The Internet, national banking network, airline reservation system, mobile phone network	Connecting networks across countries and continents

More Examples with Use Cases:

- **PAN:** A person using Bluetooth headphones connected to their Phone via Bluetooth — use case: listening to music while walking.
- **LAN:** A family connecting their laptop, printer, and smart TV to a home Wi-Fi router — use case: sharing a printer and streaming movies.
- **MAN:** A city's traffic monitoring system where cameras across the city send data to a central control room — use case: traffic management.
- **WAN:** A multinational company connecting its offices in India, USA, and UK — use case: sharing company data and video conferencing across continents.

Part 2: Network Devices

Network devices are hardware components used to connect computers and other devices to form a network.

Device	Description
Modem (MOdulator-DEModulator)	A device that converts digital signals from a computer into analog signals (modulation) for transmission over telephone lines, and converts incoming analog signals back into digital signals (demodulation).
Ethernet Card (NIC — Network Interface Card)	A hardware component installed in a computer that allows it to connect to a network. It provides the physical connection (e.g., for an Ethernet cable) and has a unique MAC address.
RJ45 (Registered Jack 45)	The standard connector used for Ethernet cables (like Cat5/Cat6). It looks like a slightly larger telephone connector.
Repeater	A device that regenerates and amplifies a weak or degraded signal over a long distance, allowing it to travel further without losing quality.
Hub	A basic networking device that connects multiple devices in a LAN. It is a **dumb** device that broadcasts any data it receives to all connected devices, leading to unnecessary traffic and collisions.
Switch	An intelligent networking device that connects multiple devices in a LAN. It learns the MAC addresses of connected devices and forwards data only to the intended recipient, making it more efficient than a hub.
Router	A device that connects different networks together (e.g., your home LAN to the Internet). It uses IP addresses to determine the best path for data packets to travel.
Gateway	A device that acts as a “gate” between two networks that use different protocols. It translates data from one protocol to another, allowing dissimilar networks to communicate.
Wi-Fi Card	A wireless network interface controller that allows a device to connect to a network using Wi-Fi (Wireless Fidelity).

MAC Address (Media Access Control Address)

MAC Address stands for **Media Access Control Address**. It is a unique 48-bit (6-byte) hardware address assigned to a Network Interface Card (NIC) by the manufacturer. It is also known as a **physical address** or **hardware address**.

Examples of MAC Addresses:

- `00:1A:2B:3C:4D:5E` — A typical Ethernet NIC
- `A4:5E:60:12:34:56` — A Wi-Fi adapter
- `00:50:56:C0:00:08` — A VMware virtual machine NIC (note: VMware uses its own OUI)
- `3C:5A:B4:9F:1D:22` — A Bluetooth adapter

MAC vs IP Address: A MAC address is a permanent physical address (like a fingerprint), while an IP address is a logical address that can change (like a postal address). MAC works at Layer 2, IP works at Layer 3.

Placement of a Repeater

Cable Type	Maximum Segment Length	Purpose of Repeater
Twisted Pair (UTP)	**100 metres**	Used to regenerate the signal and extend the network beyond 100 m

Remember: In standard Ethernet using twisted-pair cable, a cable segment can be up to **100 metres**. A repeater is used when the network needs to extend beyond this distance.

Difference between Hub and Switch, Router and Gateway

a) Hub vs Switch

Feature	Hub	Switch
Function	Connects multiple devices in a network	Connects multiple devices and forwards data to the intended device

Feature	Hub	Switch
Data Transmission	Sends data to all connected devices	Sends data only to the intended device
Efficiency	Less efficient	More efficient
Security	Low — all devices see all traffic	High — traffic is isolated
Cost	Cheaper	More expensive
Usage	Used in older or simple networks	Commonly used in LANs such as school and office networks

Example Scenario: In a school computer lab with 30 computers, if a **Hub** is used, when Computer A sends data to Computer B, all 30 computers receive the data. This wastes bandwidth and causes collisions. If a **Switch** is used, the data goes only to Computer B, making the network faster and more secure.

b) Router vs Gateway

Feature	Router	Gateway
Function	Connects different networks and routes data between them	Connects networks that may use different communication protocols
Main Purpose	Determines the appropriate path for data packets	Acts as a connection point between different networks or systems
Complexity	Less complex	More complex
Speed	Faster	Slower (due to protocol conversion)
Example	Connects a home or school LAN to the Internet	Connects networks using different protocols

Example Scenario (Router): Your home Wi-Fi router connects your home LAN (using private IPs like 192.168.1.x) to the Internet (using public IPs). It routes packets between these two IP-based networks.

Example Scenario (Gateway): A smart TV may use a gateway to communicate between its home network and a different network or service that uses a different communication protocol.

Part 3: Network Topologies

Topology refers to the physical or logical arrangement of devices (nodes) in a network.

1. Star Topology

- **Layout:** All devices are connected to a central device, such as a hub or a switch.

Pros:

- Easy to install and manage
- If one device or cable fails, only that device is affected
- Easy to add or remove devices

Cons:

- If the central hub/switch fails, the entire network goes down
- More cable required than bus topology

Use Case: A school computer lab with 40 computers connected to a central switch — if one computer's cable is faulty, only that computer is affected.

2. Bus Topology

- **Layout:** All devices are connected to a single central cable called the "backbone" or "bus."

Pros:

- Easy to install
- Requires less cable
- Cheaper than star topology

Cons:

- If the backbone cable fails, the entire network goes down
- Difficult to troubleshoot
- High chance of data collisions

Use Case: A small cyber café with 6 computers on a tight budget.

3. Tree Topology

- **Layout:** A hybrid topology that combines characteristics of Bus and Star topologies. It consists of groups of star-configured networks connected to a linear bus backbone cable.

Pros:

- Scalable
- Easy to manage in segments
- Allows hierarchical grouping of departments

Cons:

- If the backbone cable fails, the entire network is affected
- More cabling required

Use Case: A university with separate departments (Science, Arts, Commerce) each having its own star network, all connected to a central backbone.

4. Mesh Topology

- **Layout:** Every device is connected to every other device in the network through a dedicated point-to-point link.
- **Formula:** For *n* devices, the number of links required = **n(n−1)/2** (for full mesh).

Pros:

- Highly reliable — if one link fails, data can be rerouted
- No traffic congestion (dedicated links)
- Highly secure (data travels only between intended devices)
- Easy to identify faults

Cons:

- Very expensive (lots of cable and ports required)
- Difficult to install and configure
- Impractical for large networks

Use Case: Military communication networks, backbone of the Internet, and critical banking systems where reliability is paramount.

Comparison of Topologies

Feature	Star	Bus	Tree	Mesh
Cable Required	Moderate	Least	Moderate	Most
Cost	Moderate	Low	Moderate	Very High
Reliability	Good	Poor	Good	Excellent
Troubleshooting	Easy	Difficult	Moderate	Easy

Feature	Star	Bus	Tree	Mesh
Scalability	Easy	Difficult	Easy	Difficult
Best For	Office, school, home	Small, temporary networks	Large organizations, campuses	Critical, high-reliability systems

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Part 4: Introduction to Internet, URL, WWW, and Its Applications

1. Introduction to Internet

> The **Internet** is a global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a "network of networks."

- The Internet grew out of ARPANET (1969) and NSFNET (1986).
- The adoption of **TCP/IP** on January 1, 1983, is considered the birth of the modern Internet.
- The invention of the **World Wide Web (WWW)** by Tim Berners-Lee in 1989 made the Internet easily accessible to the general public.

2. URL (Uniform Resource Locator)

> A **URL** is the complete address of a resource on the web. It specifies the location of a web page, image, video, or other resource on the Internet.

Structure of a URL:

`https://www.example.com/index.html`

- **Protocol:** `https://` — the method used to access the resource.
- **Domain Name:** `www.example.com` — the human-readable address of the web server.
- **Path:** `/index.html` — the specific file or page on the server.

> **Examples of URLs:**
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> - `https://www.google.com/search?q=class12`
> - `https://www.youtube.com/watch?v=abc123`

- `https://www.cbse.gov.in/cbsenew/cbse.html`

3. WWW (World Wide Web)

The **World Wide Web (WWW)** is an information system where documents and other web resources are identified by URLs, interlinked by hypertext links, and can be accessed via the Internet. It is a service that runs *on* the Internet.

Internet vs WWW: The Internet is the physical infrastructure (hardware, cables, servers), while the WWW is a service (software) that runs on the Internet. The Internet is like the roads; the WWW is like the vehicles and buildings on those roads.

4. Applications of the Internet

a) Web (World Wide Web)

- Browsing websites, reading news, watching videos, online shopping, e-learning.
- **Examples:** Google, Wikipedia, YouTube, Amazon, Flipkart, CBSE website.

b) Email (Electronic Mail)

- Sending and receiving messages electronically over the Internet.
- Can include text, images, documents, and other attachments.
- **Examples:** Gmail, Yahoo Mail, Outlook, Rediffmail.
- **Use Case:** A student emails an assignment to a teacher; a company sends invoices to clients.

c) Chat (Instant Messaging)

- Real-time text-based communication between two or more users over the Internet.
- **Examples:** WhatsApp, Telegram, Instagram Direct, Facebook Messenger, Google Chat.
- **Use Case:** Friends chatting in a group; customer support chat on a shopping website.

d) VoIP (Voice over Internet Protocol)

- A technology that allows voice calls to be made using a broadband Internet connection instead of a regular phone line.
- Can also support video calls and conference calls.
- **Examples:** WhatsApp voice call, Skype, Zoom, Google Meet, Microsoft Teams.

- **Use Case:** A student attending an online class via Zoom; a family making an international call via WhatsApp.

Part 5: Website

1. Introduction to Website

> A **website** is a collection of related web pages and other digital assets (images, videos, documents) that are typically identified with a common domain name and published on at least one web server.

- **Example:** `www.cbse.gov.in` (CBSE official website), `www.flipkart.com`.
- A website is accessed through a **web browser** by typing its URL.

2. Difference between a Website and a Webpage

Feature	Website	Webpage
Definition	A collection of related web pages	A single document on the web
Scope	Larger — contains many web pages	Smaller — a single page within a website
URL	Has a common domain name (e.g., `www.example.com`)	Has a specific URL including the path (e.g., `www.example.com/about.html`)
Dependency	Independent — can exist on its own	Dependent — is a part of a website
Example	`www.google.com` (the entire Google site)	`www.google.com/search` (a single page)

> **Analogy:** A **website** is like a book, and a **webpage** is like a single page in that book. A book (website) contains many pages (webpages), all bound together under one title (domain name).

3. Static vs Dynamic Web Page

Feature	Static Web Page	Dynamic Web Page
Content	Fixed content — same for every user	Content changes based on user, time, or other factors
Server-Side Processing	No server-side processing needed	Requires server-side processing (PHP, ASP.NET, JSP, Python)
Database	Does not use a database	Typically uses a database to fetch content
Loading Speed	Loads faster	Loads slower (due to server processing)
Maintenance	Difficult to update — must edit HTML manually	Easy to update — content managed via CMS or database
Interactivity	Low — no user interaction beyond links	High — forms, login, search, comments
Technology	HTML, CSS	HTML, CSS, JavaScript, PHP, ASP.NET, JSP, Python, databases
Example	A personal portfolio page, an "About Us" page	Facebook feed, Gmail inbox, Amazon search results, online banking

Examples:

- **Static:** A college's "Contact Us" page that displays the same address and phone number to every visitor.
- **Dynamic:** A student logging into the CBSE results website and seeing their own marks — the page is generated dynamically based on the roll number entered.

4. Web Server and Hosting of a Website

Web Server:

A **web server** is a computer system that hosts websites and delivers web pages to clients (web browsers) upon request.

- It stores website files (HTML, CSS, images) and responds to HTTP/HTTPS requests from browsers.

- **Examples:** Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed.

Web Hosting:

> **Web hosting** is a service that provides the storage space and technology needed for a website to be accessible on the Internet.

- A web hosting company rents out server space to individuals and organizations.
- **Examples:** GoDaddy, Bluehost, HostGator, AWS (Amazon Web Services), GitHub Pages.

> **How it works:** When you type `https://www.cbse.gov.in` in your browser, the request goes to the **web server** hosting the CBSE **website**. The server processes the request and sends back the web pages, which the browser displays. The website is made available through a **web hosting** service.

Part 6: Web Browsers

1. Introduction to Web Browsers

> A **web browser** is a software application used to access, retrieve, and view websites and web pages on the Internet.

- It sends requests to web servers and renders the received HTML/CSS/JavaScript into a visual page.
- It interprets URLs and displays the corresponding web content.

2. Commonly Used Browsers

Browser	Developed By	Platform	Key Feature
Google Chrome	Google	Windows, macOS, Linux, Android, iOS	Most popular; fast; large extension library
Mozilla Firefox	Mozilla Foundation	Windows, macOS, Linux, Android, iOS	Open-source; strong privacy features
Microsoft Edge	Microsoft	Windows, macOS, Android, iOS	Default browser on Windows; built on Chromium

Browser	Developed By	Platform	Key Feature
Safari	Apple	macOS, iOS	Default browser on Apple devices; energy-efficient
Opera	Opera Software	Windows, macOS, Linux, Android, iOS	Built-in VPN; ad blocker; battery saver
Brave	Brave Software	Windows, macOS, Linux, Android, iOS	Privacy-focused; blocks ads and trackers by default

3. Browser Settings

Browser settings allow users to customize the browser's behavior and appearance. Common settings include:

- **Homepage:** The page that opens when the browser starts.
- **Default Search Engine:** The search engine used when typing in the address bar (e.g., Google, Bing, DuckDuckGo).
- **Privacy and Security:** Managing cookies, clearing browsing data, blocking pop-ups, enabling safe browsing.
- **Downloads:** Setting the default download location and asking where to save each file.
- **Appearance:** Changing theme, font size, and zoom level.
- **Languages:** Setting preferred languages for web pages.
- **Extensions/Add-ons:** Managing installed extensions.
- **Passwords:** Saving and managing login credentials.
- **Autofill:** Automatically filling forms with saved information.

> **Example:** In Google Chrome, go to `Settings → Privacy and Security → Cookies and other site data` to manage cookie settings. Go to `Settings → Appearance` to change the theme to dark mode.

4. Add-ons and Plug-ins

Add-ons (Extensions):

> **Add-ons** (also called **extensions**) are small software programs that add new features or functionality to a web browser.

- They are installed from the browser's official store (e.g., Chrome Web Store, Firefox Add-ons).
- They can modify the browser's UI, block ads, manage passwords, etc.
- **Examples:** AdBlock Plus, Grammarly, LastPass, Dark Reader, uBlock Origin.

Plug-ins:

> **Plug-ins** are software components that add specific capabilities to a web browser, typically for handling particular types of content (e.g., PDF, Flash, Java).

- They are usually third-party programs that integrate with the browser.
- Modern browsers have largely replaced the need for plug-ins by integrating features natively.
- **Examples:** Adobe Flash Player (discontinued), Java Plug-in (deprecated), PDF viewer plug-ins.

> **Add-on vs Plug-in:** An **add-on** enhances the browser's functionality (e.g., ad blocker), while a **plug-in** enables the browser to display or process specific types of content (e.g., PDF, Flash). In modern browsers, the line between them has blurred, and the term "extension" is more commonly used.

5. Cookies

> **Cookies** are small text files stored on a user's computer by a website. They contain data about the user's browsing activity and preferences.

Purpose of Cookies:

- **Session Management:** Keeping a user logged in as they navigate between pages.
- **Personalization:** Remembering user preferences (e.g., language, theme, location).
- **Tracking:** Recording browsing behavior for analytics and targeted advertising.
- **Shopping Carts:** Remembering items added to a cart on an e-commerce site.

Types of Cookies:

Type	Description	Example
Session Cookies	Temporary; deleted when the browser is closed	Keeping you logged in while navigating a website

Type	Description	Example
Persistent Cookies	Stored for a fixed period; remain after browser is closed	Remembering your login credentials for a website
First-Party Cookies	Set by the website you are visiting	Amazon remembering your shopping cart
Third-Party Cookies	Set by a domain other than the one you are visiting	Advertising networks tracking your browsing across sites

Examples:

- When you log into Gmail and check "Remember me," a persistent cookie is stored so you don’t have to log in again.
- When you add items to your Amazon cart and close the browser, a cookie remembers your cart items when you return.
- When you search for a product on Flipkart, third-party cookies may show you ads for similar products on other websites.

Privacy Concern: Cookies can be used to track user behavior across websites, raising privacy concerns. Users can manage or delete cookies through browser settings. Many browsers now block third-party cookies by default.